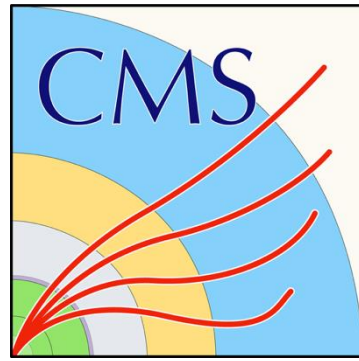




PRINCETON
UNIVERSITY

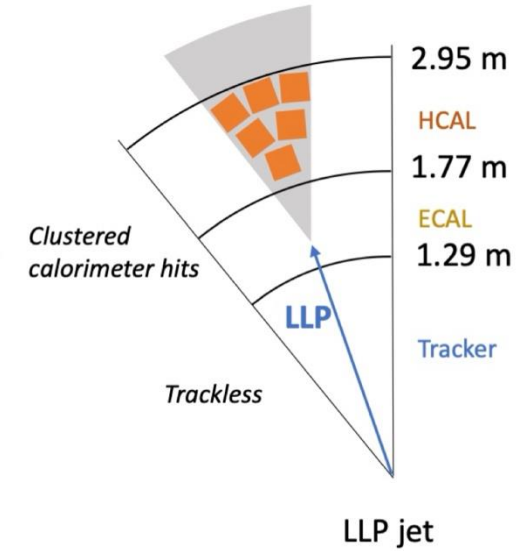
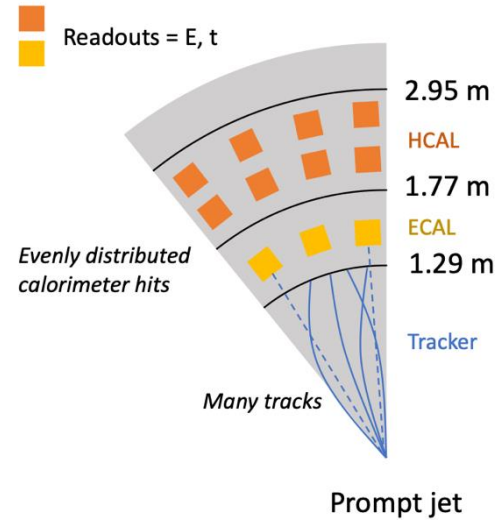
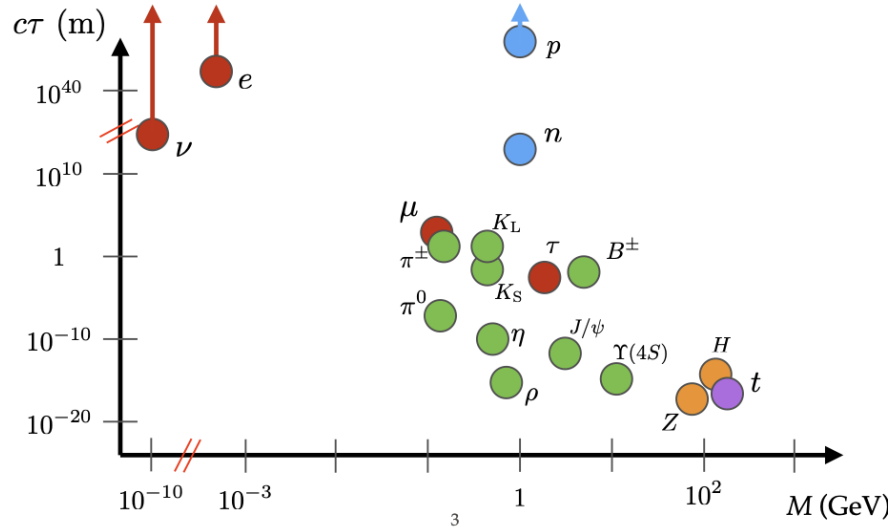
Caltech



Search for Long-Lived Particles with HCAL Segmentation in CMS at the Large Hadron Collider

Katia Avanesov, Harvey Newman, Kiley Kennedy, Gillian Kopp, Cristian Peña, Chris Tully, Si Xie
Taylor Lawrence SURF Research Fellow

Long Lived Particles (LLPs)

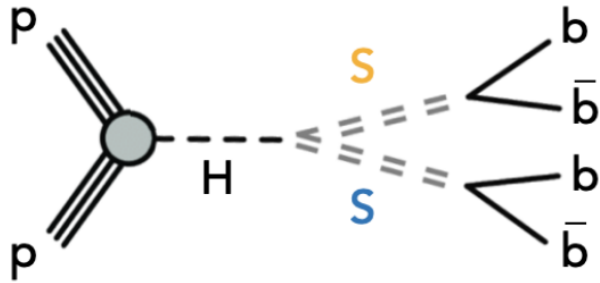


- LLPs have lifetimes greater than 0.1 nanoseconds
- LLPs feature in both the Standard Model and predicted in numerous Beyond the Standard Model (BSM) theories

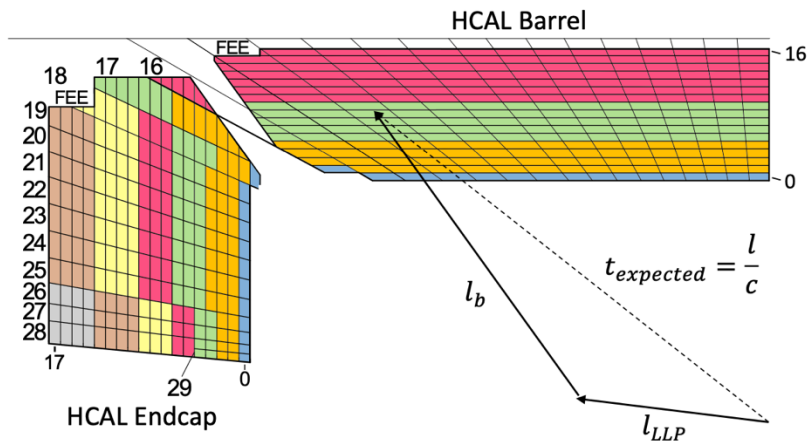
- Trackless due to decays in the HCAL
- Displaced vertex
- Narrower energy spread
- Longer lived particles will deposit less energy in the earlier layers of HCAL
- Delayed due to longer lifetime and path length

Search Strategy

- Searching for $H \rightarrow ss \rightarrow bbbb$



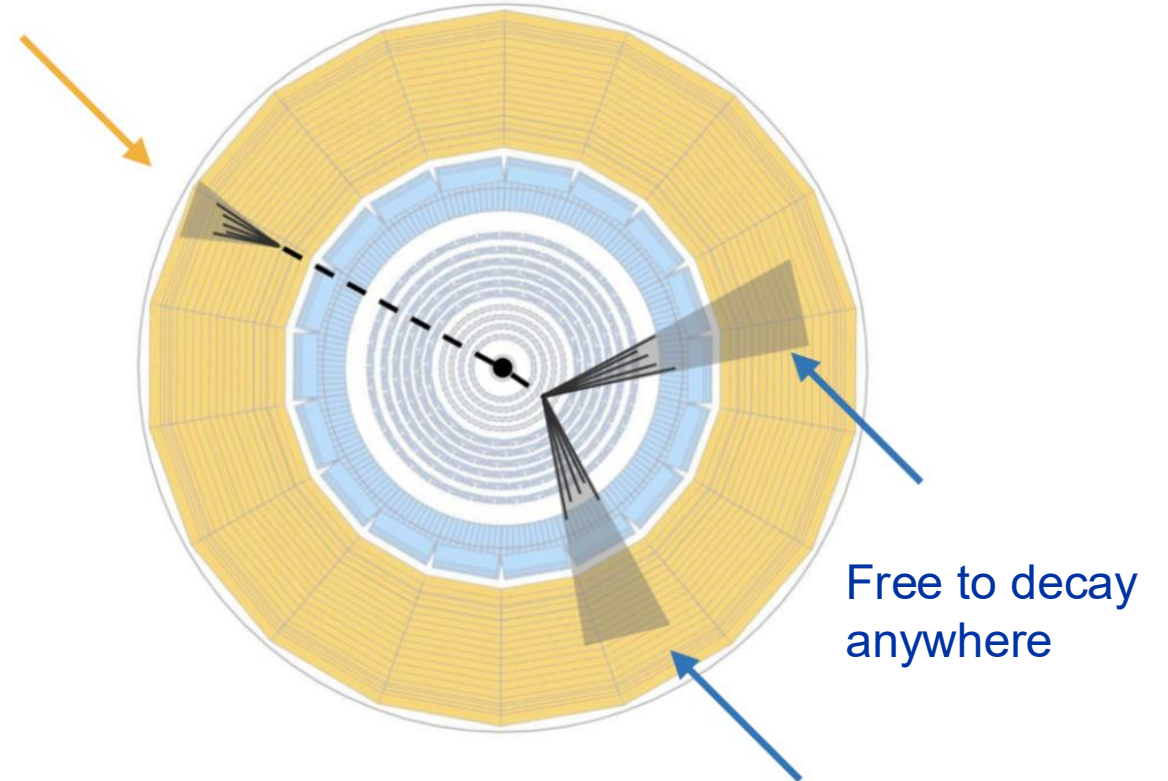
- Use 4 layer HCAL depth segmentation



$$\Delta t = \frac{l_{LLP}}{v_{LLP}} + \frac{l_b}{c} - t_{expected}$$

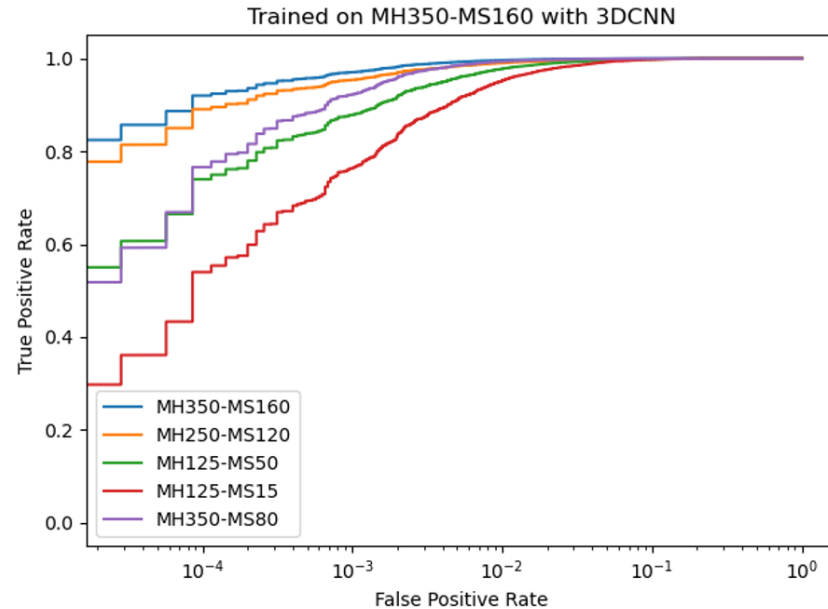
- Event Selection: at least one LLP decays in HCAL

Decays in HCAL

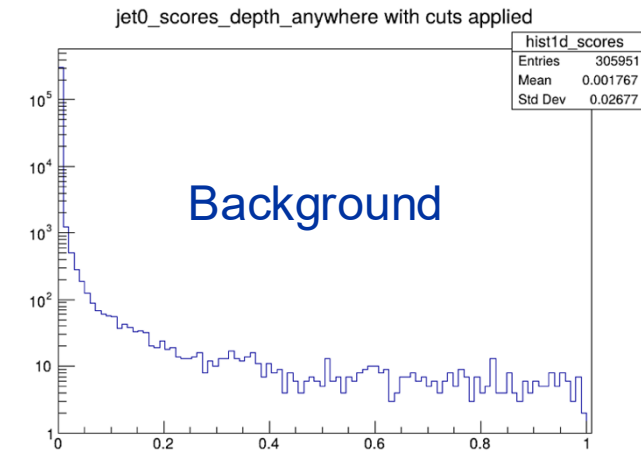
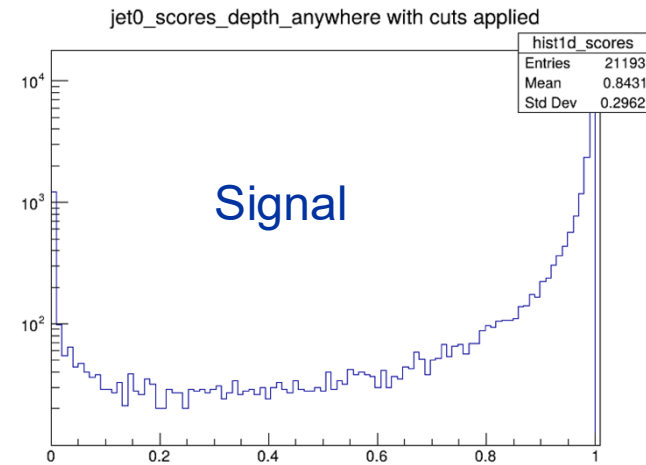


Analysis Strategy

- Train two distinct DNNs and evaluate them on the two LLPs in each event



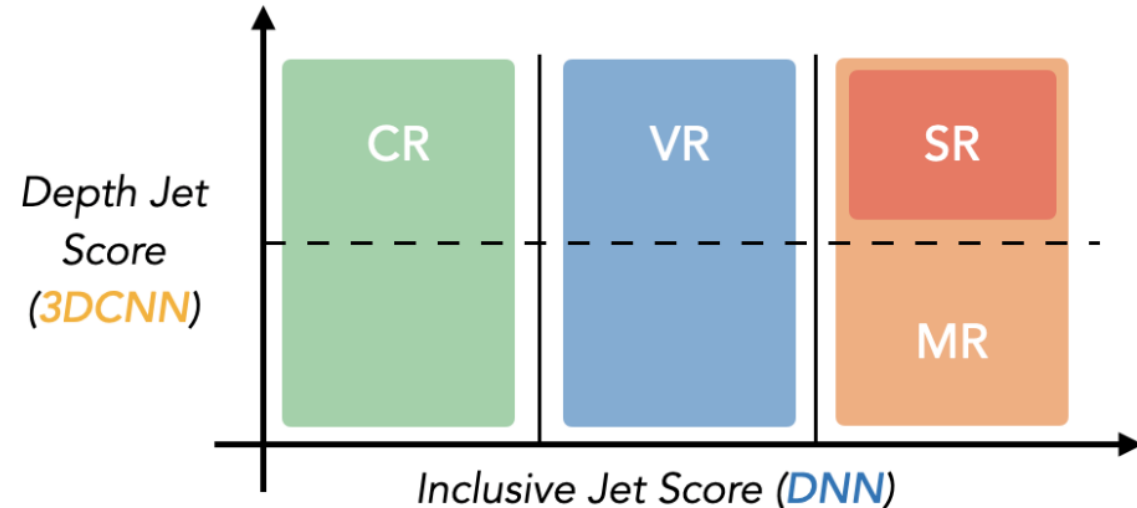
- Perform a counting experiment



Region Definition

Categorize events by inclusive DNN score:

Control, Validation, Measurement, Signal



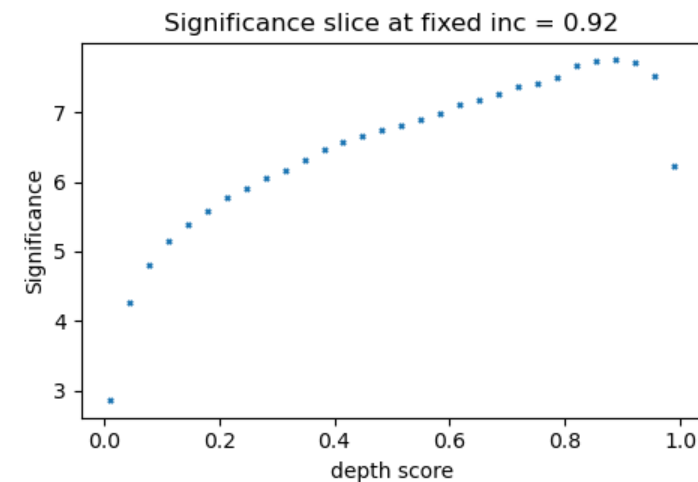
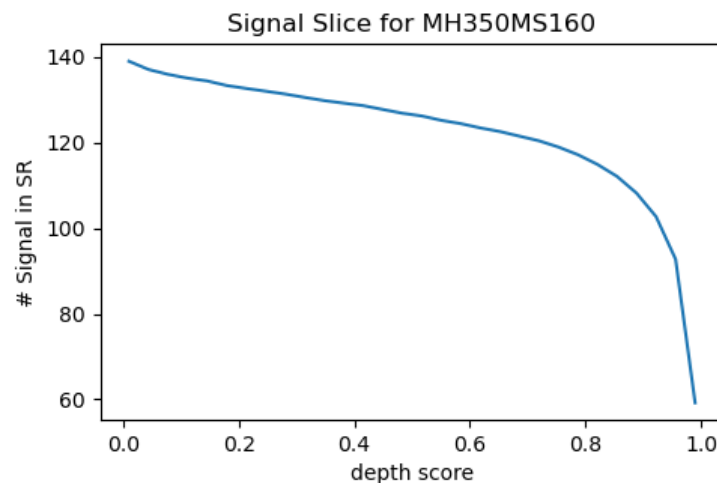
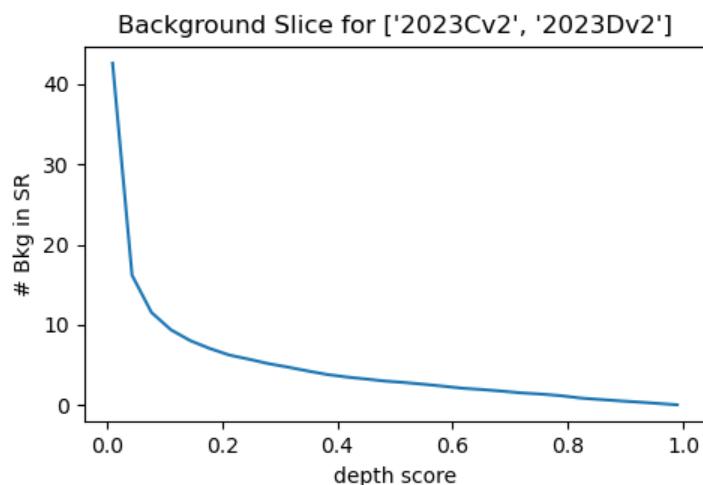
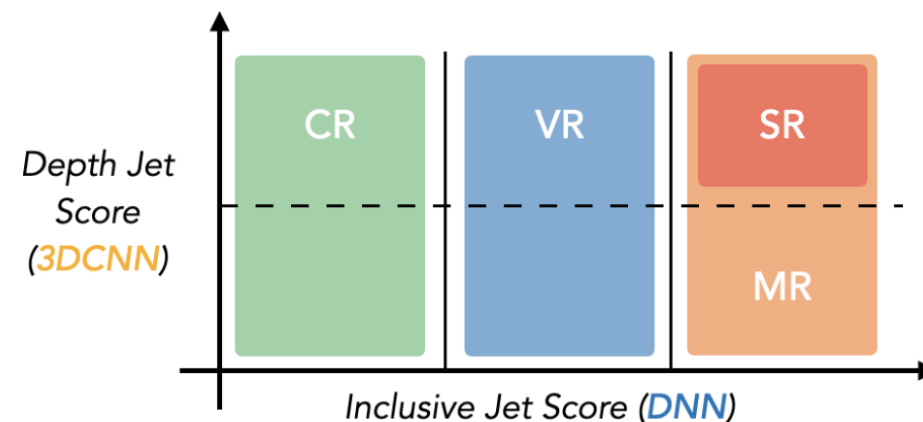
Optimizing Signal Region Definition

Region Definition

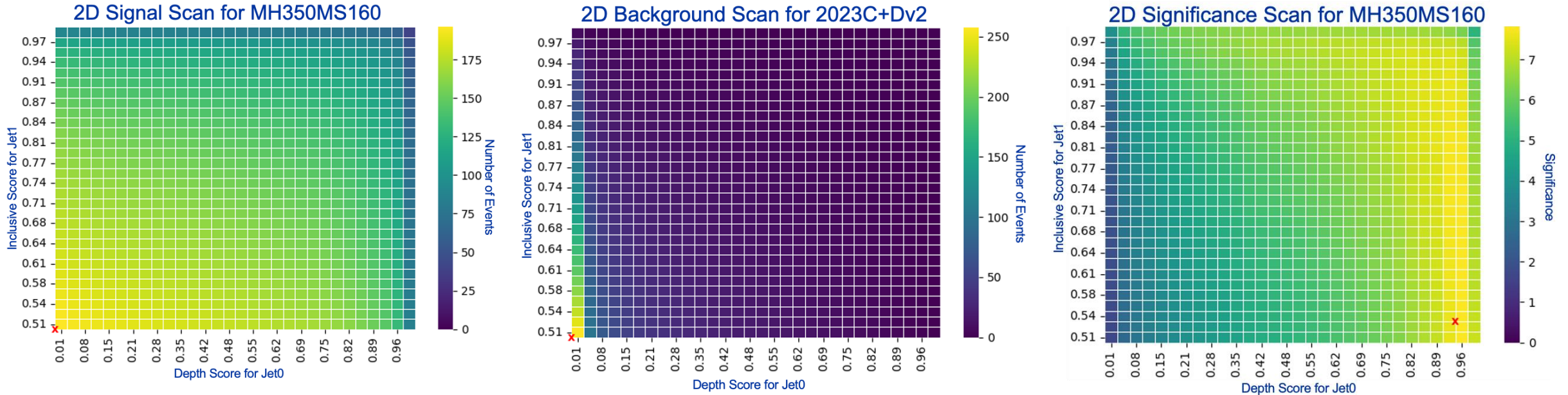
Categorize events by inclusive DNN score:

Control, Validation, Measurement, Signal

- How can we define our signal region to maximize sensitivity to counting excesses?
- What values of (inclusive score, depth score) achieve maximal sensitivity?

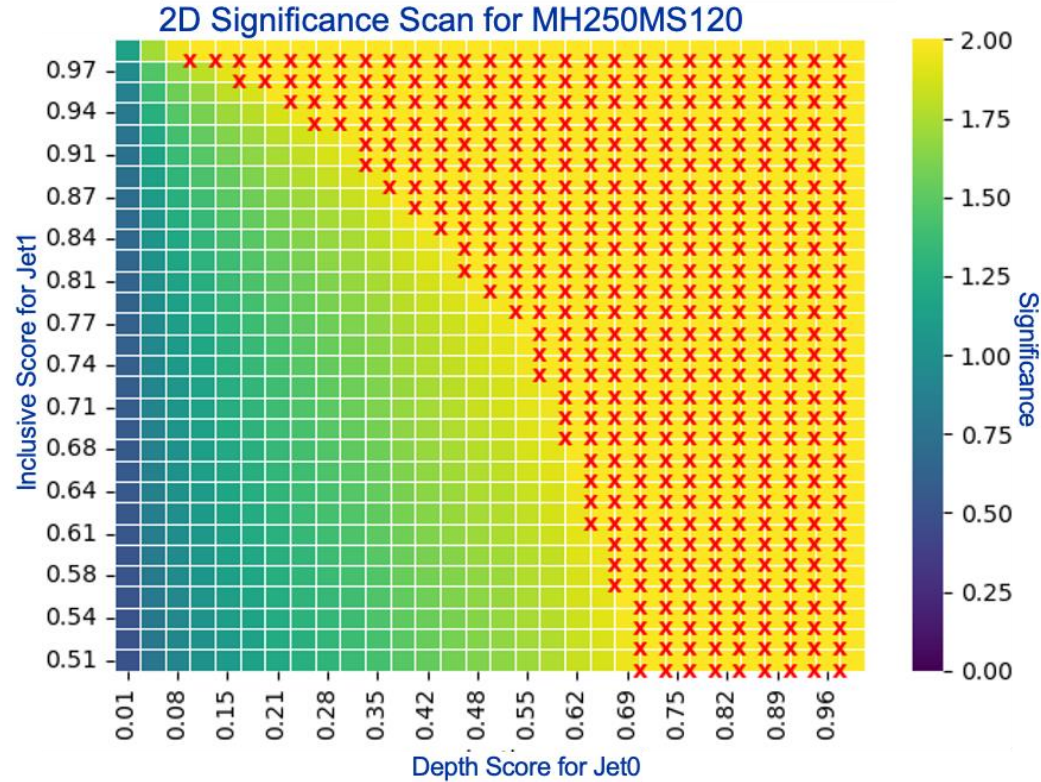


Signal Region Optimization in 2D

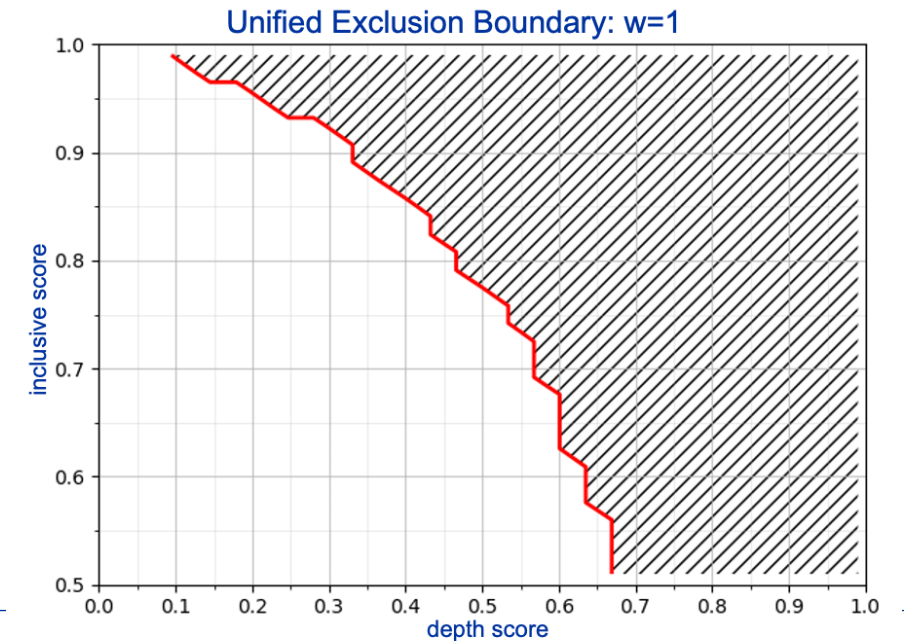
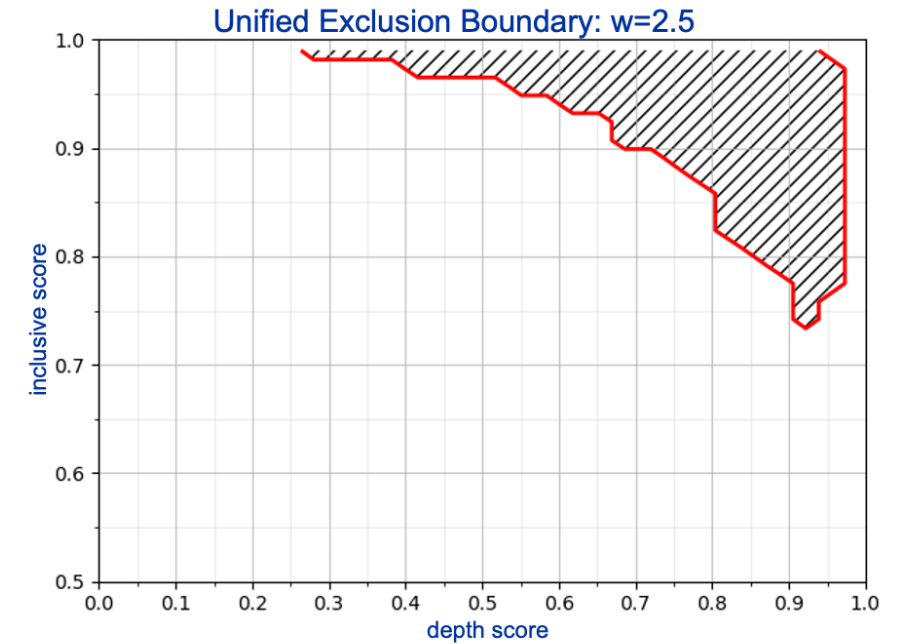


- Scan the threshold space and calculate significance
- Perform scans across various simulations: varying Higgs Mass, LLP mass, triggering on leading jet, estimated branching ratio
- Require significance > 2 for statistically meaningful results

Unifying Exclusion Boundaries



- To visualize the set of thresholds which exclude as many possible processes, we take the intersection of generated exclusion boundaries



Conclusion

- **Building on previous work: integrating the NNs into a formal analysis framework and strategy**
- **Progress towards optimizing the final threshold selection before unblinding the analysis**

THANK YOU!

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